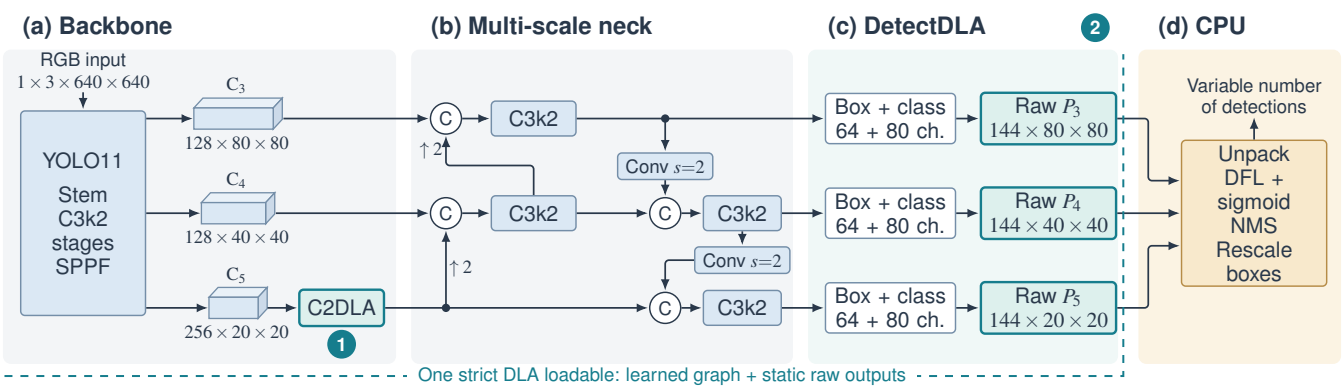
(a) Backbone
(b) Multi-scale neck
(c) DetectDLA
2
(d) CPU
RGB input
1 × 3 × 640 × 640
YOLO11
Stem
C3k2
stages
SPPF
C3
128 × 80 × 80
C4
128 × 40 × 40
C5
256 × 20 × 20
C2DLA
1
C
C3k2
↑ 2
Conv s=2
Box + class
64 + 80 ch.
Raw P3
144 × 80 × 80
Raw P4
144 × 40 × 40
Raw P5
144 × 20 × 20
Variable number
of detections
Unpack
DFL +
sigmoid
NMS
Rescale
boxes
One strict DLA loadable: learned graph + static raw outputs
1 C2PSA → C2DLA
2 raw export boundary
C: channel concat
↑2: nearest upsampling
Dimensions omit batch B=1